

Lung Nodule Classification Project Overview

1. This project builds an end-to-end, multimodal machine learning pipeline to classify lung nodules as Benign or Malignant.
2. It uses a [Hybrid Gated Fusion Model](#) that dynamically merges visual data (3D CT scans) and tabular data (clinical/radiomic features) to output a highly accurate consensus score.

1. Data Acquisition

- **File:** [downloader.ipynb](#)
- **Function:** Connects to The Cancer Imaging Archive (TCIA) using the `tcia_utils` library to download the raw LIDC-IDRI DICOM CT scans directly into Google Drive (1,018 series total).

2. Data Preprocessing & Feature Extraction

- **File:** [phase1preprocess.ipynb](#)
- **Function:** Translates raw medical images into a standardized format for the AI.
 - Uses `SimpleITK` and `pydicom` to resample volumes to an isotropic 1.0mm spacing.
 - Applies a lung window filter (-1000 to 400 HU) to isolate lung tissue.
 - Extracts uniform 64x64x64 3D patches around the centroid of each nodule.
 - Calculates the consensus malignancy score and compiles radiomic features (subtlety, sphericity, margin, spiculation, texture) into the master `manifest.csv` file.

3. Model Architecture Prototyping

- **File:** [phase2.ipynb](#)
- **Function:** Defines the PyTorch Dataset (`LungNoduleDataset`) and the core `HybridGatedFusionModel`.
 - The model uses a `CNN3DBranch` for the 3D images and a `TabularMLPBranch` for the clinical data, merged by a Gated Fusion Layer.
 - Sets up a rapid 5-fold cross-validation loop (`GroupKFold`) on a smaller subset of data to verify the architecture can learn without data leakage.

4. Evaluation & Explainable AI (XAI)

- **File:** [phase3_evaluation.ipynb](#)
- **Function:** Evaluates the model's clinical viability.
 - Runs ensemble inference across the 5 models, achieving an impressive ROC-AUC score of 0.9042.

- Utilizes Youden's J Statistic to calculate the mathematically optimal probability threshold (57.83%) for diagnosis.
- Defines the **GradCAM3D** class to generate visual heatmaps of the model's focal points.
- Executes a Modality Ablation Study to prove that both the 3D CNN and Tabular MLP branches are necessary for peak accuracy.

5. Full Cohort Optimized Training

- **File:** [phase4_full_scale.ipynb](#)
- **Function:** Handles the heavy computational training on the complete dataset (1,242 nodules).
 - Applies 3D spatial augmentations (**RandomFlip**, **RandomAffine**) using the **torchio** library to prevent overfitting.
 - Utilizes Automatic Mixed Precision (AMP) to optimize GPU memory and speed during training.
 - Implements class weights within the **CrossEntropyLoss** function to account for the imbalance between benign and malignant samples.

6. Live Diagnostic Dashboard

- **File:** [final_inference_pipeline.ipynb](#)
- **Function:**
 - The final deployment script that loads the trained ensemble weights into a live interactive web application using **gradio**.
 - It allows a user to select specific nodules, adjust clinical features, and receive real-time predictions alongside the 3D Grad-CAM heatmaps.

II. Architecture & Evaluation

The second and third phases define the brain of the AI and prove its clinical viability:

- **Hybrid Architecture:** A 3D CNN branch processes the image tensors, while a Tabular MLP processes the radiomic scores.
- **Optimal Calibration:** Uses Youden's J Statistic to mathematically determine the best probability threshold (57.83%) for diagnosis.
- **Explainable AI (XAI):** Implements [3D Grad-CAM](#) to generate visual heatmaps that show exactly which anatomical features influenced the model's decision.

III. Full Scale Training & Deployment

The final steps scale the model for production:

- **Optimized Training:** Leverages [torchio](#) for 3D augmentations and Automatic Mixed Precision (AMP) to train across all nodules efficiently without overfitting.
- **Interactive Dashboard:** Wraps the best pre-trained weights into a live [gradio](#) web application, allowing users to select nodules, tweak features, and view immediate AI predictions.

A Detailed breakdown of the core Machine Learning (ML) and Deep Learning (DL) concepts implemented across your project files.

1. Multimodal Learning & Gated Fusion Architecture

- **Multimodal Representation:** Instead of relying solely on images (unstructured data) or radiomic scores (structured tabular data), the model learns from both modalities simultaneously.
- **3D Convolutional Neural Network (CNN):** Operates directly on volumetric spatial data $(C, D, H, W) = (1, 64, 64, 64)$ to extract 128 high-level morphological features without flattening spatial depth.
- **Multi-Layer Perceptron (MLP):** A sub-network that maps the 5 continuous clinical/radiomic features into a 32-dimensional embedding.
- **Gated Attention Mechanism:** After concatenating the 128 visual features and 32 tabular features (160 total), a sigmoid gating layer learns dynamic importance weights:

$$\bullet \quad z_{gated} = x \odot \sigma(Wx + b)$$

- This allows the network to suppress noisy image features if the clinical features are clearer, or vice versa.

2. Generalization & Data Leakage Prevention

- **GroupKFold Cross-Validation:** In medical datasets, a single patient often has multiple nodules. Splitting data randomly causes **data leakage** because scans from the same patient share scanner artifacts and biological traits across train/validation sets. Grouping splits by `patient_id` ensures zero patient overlap between folds.
- **Per-Fold Feature Standardization:** Tabular data normalization (`StandardScaler`) is fitted strictly on the training partition of each fold before transforming the validation partition, preventing future distribution leakage.
- **3D Spatial Data Augmentation:** Using `torchio` to apply random 3D rotations (`RandomAffine`) and spatial flipping (`RandomFlip`) to synthetically expand the dataset and prevent the 3D CNN from memorizing slice orientations.

3. Optimization & Imbalanced Classification

- **Class-Weighted Loss:** Benign nodules heavily outnumber malignant ones. To prevent the model from biasing toward the majority class, the loss function uses inverse class frequency weights:

$$\bullet \quad w_c = \frac{1}{count(c)}$$

- **Automatic Mixed Precision (AMP):** Utilizes `torch.cuda.amp` (`autocast` and `GradScaler`) to compute gradients in 16-bit floating point (`float16`) where possible while retaining 32-bit (`float32`) master weights, doubling training speed and cutting VRAM usage.
- **Cosine Annealing Schedule & AdamW:** Uses `AdamW` (Adam with decoupled weight decay for L_2 regularization) combined with `CosineAnnealingLR` to smoothly decay the learning rate over 25 epochs.

4. Post-Hoc Evaluation, Ensembling & Explainable AI (XAI)

- **Model Ensembling (Soft Voting):** Averages predicted class probabilities across all 5 cross-validation models, reducing prediction variance and boosting overall ROC-AUC (0.9042).
- **Threshold Calibration (Youden's J Statistic):** The default 0.50 classification threshold is suboptimal for imbalanced medical data. Phase 3 optimizes the cutoff point using Youden's J index:
 - $J = \text{Sensitivity} + \text{Specificity} - 1 = \text{TPR} - \text{FPR}$
- This identified 0.5783 as the optimal threshold for maximizing malignant detection sensitivity while minimizing false alarms.
- **3D Grad-CAM (Gradient-Weighted Class Activation Mapping):** Uses PyTorch forward and backward hooks on the final 3D convolution layer to compute gradients with respect to the malignant class score, producing 3D heatmap overlays that visually highlight what influenced the AI's diagnosis.

FUTURE WORKS:

1. Validate on External Datasets

Currently, your model is evaluated solely on the LIDC-IDRI dataset.

A recognized limitation in state-of-the-art medical imaging research is single-database validation.

To prove that your model generalizes well to real-world clinical settings without bias, consider testing your pre-trained weights on external lung nodule databases like the [ELCAP](#) or NELSON datasets.

2. Advanced Data Augmentation

While you are currently using spatial augmentations (rotations and flips), you are still dealing with an imbalanced dataset where benign cases outnumber malignant ones.

Advanced researchers often use Generative Adversarial Networks (GANs) to synthesize artificial images of the rare malignant classes to create a perfectly balanced training set.

3. Track State-of-the-Art Benchmarks

Your current ensemble model achieved a very impressive ROC-AUC score of 0.9042.

However, [state-of-the-art methods](#) for this specific LIDC-IDRI classification task are currently pushing between 0.94 and 0.97.

Tracking these benchmarks can inspire you to experiment with newer architectures, like Multi-Granularity Transformers.

4. Ensure Safety in Deployment Calibration

If you attempt to make your live dashboard faster by compressing or quantizing the model, standard accuracy metrics are no longer enough.

You must evaluate safety-oriented indicators—such as the Expected Calibration Error (ECE)—to ensure the model doesn't become overly confident when it makes incorrect predictions on compressed data.

Maintaining high-quality [Explainability](#) (like your Grad-CAM heatmaps) is a core requirement for validating these deployment-ready clinical models.